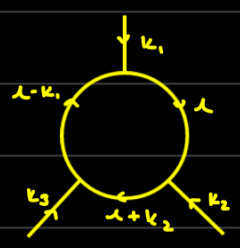
Loop Corrections To the Vertex

Solving w. Regularization

The diagram shown on the right is the correction to the ϕ^3 vertex. We define a vertex function such that $k_1+k_2+k_3=0$ and it is a sum of corrections:

can be -ve to denote outgoing nature

$$i\vec{V}_3(k_1,k_2,k_3) = \underbrace{iZ_g g}_{\text{The normal vertex}} + (ig)^3\left(\frac{1}{i}\right)^3\underbrace{\int\frac{d^d l}{(2\pi)^d}\,\tilde{\delta}((l-k_1)^2)\,\tilde{\delta}(l^2)\,\tilde{\delta}((l+k_2)^2)}_{\text{One-Loop corrections}} + O(g^5)$$



We can use Feynman's method to simplify this:

$$\tilde{\delta}((l-k_1)^2)\,\tilde{\delta}(l^2)\,\tilde{\delta}((l+k_2)^2) = \int dF_3\left[x_1(l-k_1)^2 + x_2(l+k_2)^2 + x_3 l^2 + m^2\right]^{-3}$$

$$= \int dF_3\left[l^2 - 2l(x_1k_1 - x_2k_2) + x_1k_1^2 + x_2k_2^2 + m^2\right]^{-3} = \int dF_3\left[q^2+D\right]^{-3}$$

Here, we have defined: $q = l - x_1k_1 + x_2k_2$; $D = x_3x_1k_1^2 + x_3x_2k_2^2 + x_1x_2k_3^2 + m^2$

Simplified using $\delta(x_1+x_2+x_3-1)=0$

We can make a Wick rotation of the q^0 contour to get:

$$\frac{1}{g}\vec{V}_3(k_1,k_2,k_3) = Z_g + g^2\int dF_3\int\frac{d^d\bar{q}}{(2\pi)^3}\frac{1}{(\bar{q}^2+D)^3} + O(g^4)$$

$\longrightarrow$ Diverges for $d\geq 6$ and thus we can implement Dimensional Regularization.

Q: What exactly is the vertex function? Is it a vector function? What does it exactly represent? Is this function like a substitute for $iZ_g g$, so $iZ_g g \longrightarrow iV_3$ in loop diagrams.

A:

We can express this integral in terms of Euler-gamma functions to be:

$$\int\frac{d^d\bar{q}}{(2\pi)^d}\frac{1}{(\bar{q}^2+D)^3} = \frac{\Gamma(3-\frac{1}{2}d)\,D^{-(3-d/2)}}{2(4\pi)^{d/2}}$$

Now if we do the usual: $d=6-\epsilon$; $g\rightarrow g\tilde{\mu}^{\epsilon/2}$, we get:

$$\frac{1}{g}\vec{V}_3(k_1,k_2,k_3) = Z_g + \frac{1}{2}\alpha\,\Gamma\left(\frac{\epsilon}{2}\right)\int dF_3\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\epsilon/2} + O(\alpha^2)$$

$\longrightarrow$ Here, $\alpha = \frac{g^2}{(4\pi)^3}$

Taking the $\epsilon\rightarrow 0$ limit gives us:

$$\frac{1}{g}\vec{V}_3(k_1,k_2,k_3) = Z_g + \frac{1}{2}\alpha\left[\frac{2}{\epsilon} + \int dF_3\ln\left(\frac{4\pi\tilde{\mu}^2}{e^{\gamma}D}\right)\right] + O(\alpha^2)$$

If we make the substitutions: $\mu^2 = 4\pi e^{-\gamma^2}\tilde{\mu}^2$, $Z_g = 1+C$, we get:

$$\frac{1}{g}\vec{V}_3(k_1,k_2,k_3) = 1 + \left\{\alpha\left[\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right)\right] + C\right\} - \frac{1}{2}\alpha\int dF_3\ln\left(\frac{D}{m^2}\right) + O(\alpha^2)$$

Purely Numerical Constant

If we take c as: $c = -\alpha\left[\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right) + k_c\right] + O(\alpha^2)$

This gives us:

$$\frac{1}{g}\vec{V}_3(k_1, k_2, k_3) = 1 - \frac{1}{2}\alpha\int dF_3 \ln\left(\frac{\Lambda}{m^2}\right) - k_c\alpha + O(\alpha^2)$$

The integral over dF_3 cannot be done in closed form.

We can choose c to be like this as it renders $\vec{V}_3(k_1, k_2, k_3)$ finite and independent of μ as required.

To get the exact value of k_c, we would need a condition comparable to that of the exact propagator where $\Pi(-m^2) = 0$, $\Pi'(-m^2) = 0$. However, these conditions are there because of the known properties of the exact propagator, there is no such case here. Thus, we can choose any k_c. Each choice of k_c will lock in a particular value of g.

We then happily set $k_c = 0$ to get:

$\vec{V}_3(0,0,0) = g$

The vertex corrections are similar to their propagator counterparts in that they vary logarithmically with $|k_i^2|$ for $|k_i^2| \gg m^2$.